
Disentangling the Black Box: Interpretable Protein-Ligand Docking via Sparse Autoencoders

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Abstract

1 Protein-ligand docking pipelines based on variational autoencoders (VAEs) achieve
2 strong predictive performance but produce opaque latent representations, while
3 the large number of candidate poses generated per complex imposes substantial
4 computational cost as each undergoes expensive energy minimization and refine-
5 ment regardless of quality. Physics-informed generative pipelines like the Friesner
6 Group’s IFD-ML pipeline ground predictions in thermodynamic principles, making
7 learned features more likely to be physically meaningful and interpretability more
8 tractable and valuable than in end-to-end models. We apply sparse autoencoders
9 (SAEs) to the VAE latent space of the IFD-ML pipeline, training on 435,069 poses
10 across 36 protein systems. At a 4Å threshold, SAE-derived features filter 16.5% of
11 poor poses prior to refinement while retaining 91.1% of native-like conformations,
12 outperforming the pipeline’s own energy score. Fraction-of-variance-explained
13 (FVE) analysis shows the top SAE retains 62% of latent variance while still support-
14 ing a high-precision filter. Sparse features activate exclusively on failed docking
15 attempts, each corresponding to a distinct structural failure mode—an interpretable
16 window into what generative models learn about binding physics. These results
17 show SAEs can serve simultaneously as practical computational filters and inter-
18 pretability tools for building trustworthy drug-discovery pipelines.

19 1 Introduction

20 Protein-ligand binding governs essentially all biological processes and underlies the majority of
21 small-molecule drug discovery [Dhakal et al., 2022, Sim et al., 2025]. Predicting how a ligand binds
22 a flexible protein target, the induced fit docking (IFD) problem [Sherman et al., 2006], remains
23 one of computational biochemistry’s hardest tasks, addressed by physics-based pipelines such as
24 IFD-MD [Miller et al., 2021]. Most end-to-end machine learning (ML) approaches, such as DeepDTA
25 and its graph-based successors, treat binding as a sequence-to-structure mapping without explicitly
26 modeling atomic interactions [Wu et al., 2025], and are further limited by scarce training data, poor
27 interpretability, and weak generalization to novel proteins or ligands [Lam and Katritch, 2025, Jiang
28 et al., 2026].

29 Schrödinger and the Friesner Group’s IFD-ML pipeline combines physics-based simulation with
30 variational autoencoder (VAE) and diffusion generative models to sample conformational space,
31 achieving strong performance in cross-docking benchmarks similar to Mansoor et al. [2024] (Kon-
32 stantinovsky et al., in preparation). Because it relies on a VAE, each candidate pose is represented by
33 a fixed-dimensional continuous latent vector—the kind of representation a sparse autoencoder (SAE)
34 is designed to decompose into interpretable features (Section 2.2). This creates two problems. First,
35 the pipeline generates thousands of candidate poses per complex, each undergoing expensive Prime
36 energy minimization [Li et al., 2011] and algorithmic refinement regardless of quality—a bottleneck
37 at scale [Wang et al., 2019]. Second, the VAE’s latent vectors are opaque: it is unclear which
38 directions encode binding geometry, energetic favorability, or structural quality, and understanding

39 why a model scores a pose highly matters as much as the score itself [Qadri et al., 2025]. This
40 paper addresses both: a practical filter that removes likely-poor poses before refinement, and an
41 interpretability method that opens up what the latent space has learned.

42 Sparse autoencoders (SAEs) decompose opaque neural representations into interpretable, monose-
43 mantic features [Templeton et al., 2024, Gao et al., 2024]. Applied to protein language models, SAEs
44 recover biologically meaningful features corresponding to secondary structure, binding sites, and
45 evolutionary conservation [Adams et al., 2025, Simon and Zou, 2025]. Physics-informed pipelines
46 like IFD-ML are a compelling target: because the latent space is constrained by thermodynamic priors
47 and atomic interactions, learned features are more likely to correspond to real physical quantities—
48 energy terms, geometric constraints, binding physics—than in end-to-end models [Li et al., 2026, Wu
49 et al., 2024]. SAEs have been applied to structure-generation models such as RFDiffusion (FoldSAE)
50 [Zarzecki et al., 2025] and protein folding representations (ESMFold/ESM2-3B) [Parsan et al., 2025],
51 but to our knowledge not yet to a physics-informed, molecular-docking generative model.

52 This work applies TopK SAEs to the VAE latent space of the IFD-ML pipeline to serve both goals at
53 once. Concretely, we make three contributions: (1) a practical pre-refinement filter that flags 16.5% of
54 poses for removal while retaining 91.1% of native-like conformations; (2) evidence that the pipeline’s
55 own energy score cannot substitute for this filter, structurally unable to reach the SAE’s operating
56 points; (3) sparse features that isolate distinct, physically interpretable failure modes, from severe
57 rotational misorientation to milder translational displacement, in what generative models learn about
58 binding physics.

59 2 Methods

60 2.1 Dataset

61 We compiled a dataset of 435,069 molecular docking poses across 36 protein-ligand systems generated
62 by the IFD-ML pipeline, split by case (not pose) to prevent leakage. Each pose consists of a 30-
63 dimensional VAE latent vector, ligand RMSD to the native structure (excluding hydrogen atoms),
64 and a free energy prediction from physics-based scoring. We use two RMSD-based pose-quality
65 definitions: native-like at $\text{RMSD} \leq 2\text{\AA}$ (12.3% native-like, following Shim et al. [2022]), and good at
66 $\text{RMSD} \leq 4\text{\AA}$ (36.6% native-like) for the conservative filtering task. Splits are approximately 70%
67 train, 15% validation, 15% test; normalization uses the training split only, and feature/threshold/combo
68 selection for filtering (Section 2.4) draws on pooled train+val, with test reserved for a single final
69 evaluation.

70 2.2 TopK Sparse Autoencoder

71 We train TopK SAEs with $4\times$ expansion (30D input $\rightarrow$ 120D hidden $\rightarrow$ 30D reconstruction) fol-
72 lowing Adams et al. [2025]; the TopK operation retains the K largest activations and zeros the
73 rest, with reconstruction loss as the training objective. The input is mean-centered via a learned
74 pre-encoder bias, initialized to the training-set mean of the normalized activations and subtracted
75 before encoding, following Simon et al. [2026]’s dead-feature mitigation. We trained SAEs across
76 $K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds per K , for up to 100 epochs using Adam ($\text{lr} = 10^{-3}$, batch size
77 256), selecting the best run per K by validation loss; we report results for $K \in \{3, 8, 15\}$ in the main
78 text (the full sweep across all seven K values follows similar trends but is omitted here for space),
79 selecting the operating K per analysis (Section 3).

80 2.3 Pose Quality Classification

81 To assess whether SAE features provide utility beyond raw latents, we trained logistic regression
82 classifiers on (1) 30D normalized VAE latents (L2 penalty, class-balanced) and (2) 120D sparse SAE
83 activations for $K \in \{3, 8, 15, 20, 30\}$ (L1 penalty, class-balanced), fit on train and evaluated on test.
84 auPR is our primary metric, since it handles class imbalance better than ROC-AUC, which we also
85 report for comparison with prior work.

2.4 Threshold-Based Pose Filtering

The paper’s central practical result is a pre-refinement filter, built using a three-way train/select/test split so reported performance cannot be inflated by threshold or feature selection on the test set. For each of the 120 features per SAE, we fit an F1-maximizing activation cutoff and rank candidates by precision on a pooled train+val selection set, requiring recall > 0.02 to exclude degenerate near-zero-recall picks. We take the top 16 candidates, then search all subset unions—evaluated on the same selection set—for the union maximizing recall subject to retaining $\geq 80\%$ of native-like poses; the winning union is applied exactly once to test, which is never touched during fitting or selection. To confirm that precision derives from the SAE decomposition rather than raw latent geometry, we apply the identical procedure to 500 random unit directions and all 60 signed raw latent axes. Finally, as a physics-based baseline, we apply the same discipline directly to the IFD-ML pipeline’s pre-minimization energy score, accounting for its sentinel value (10,000 kcal/mol)—assigned when the ligand is not in the basin that refinement operates in, or the pose is otherwise too structurally poor to score meaningfully.

3 Results

3.1 Reconstruction, Classification, and Sparsity

Does decomposing the latent space into sparse features cost any of the predictive signal present in the raw, dense latents? Table 1 reports auPR for bad-pose classification, FVE, and dead-feature rate, all on the test set. Raw latents match or exceed every SAE K on auPR at both thresholds—the sparse decomposition provides no consistent predictive lift, so its value lies in filtering (Section 3.2) and interpretability (Section 3.4) rather than classification. FVE grows monotonically with K without a matching auPR gain, and dead-feature rate spikes to 25.8% at $K=15$ despite the mean-centering mitigation from Section 2.2 [Simon et al., 2026].

Table 1: auPR, FVE, and dead-feature rate, raw latents vs. SAE K (test set).

Rep.	auPR (4Å)	auPR (2Å)	FVE	Dead %
Raw latents	0.770	0.855	—	—
SAE $K=3$	0.709	0.858	33.7%	2.5%
SAE $K=8$	0.719	0.821	62.0%	1.7%
SAE $K=15$	0.750	0.846	90.0%	25.8%

3.2 Practical Filtering

We now evaluate the pre-refinement filter from Section 2.4. Table 2 reports the combo filter for $K \in \{3, 8, 15\}$ at both thresholds, alongside the raw-latent-geometry baselines (Section 2.4). $K=8$ and $K=15$ are the reproducible headline operating points, at 4Å and 2Å respectively ($K=3$ ’s combo cleared the 80% good_kept bar on the selection set at 85.2% but dropped to 77.5% on held-out test—a genuine generalization failure rather than a threshold artifact): $K=8$ flags 16.5% of test poses while retaining 91.1% of native-like conformations (precision 0.803); $K=15$ flags 15.7% while retaining 93.0% (precision 0.935). Recall is modest at both (0.21, 0.17)—a deliberate precision-first design choice, since retaining native-like poses matters more than catching every poor one for a pre-refinement filter.

Neither raw-latent-geometry baseline family reaches the good_kept $\geq 80\%$ target at either threshold (0 of 65,535 candidate-union combinations clear it), so Table 2 reports each baseline’s closest achievable point instead—well below what $K=8/K=15$ achieve while flagging fewer poses.

3.3 Comparison to the IFD-ML Pipeline’s Energy Score

Applying the same selection discipline (Section 2.4) to the pipeline’s pre-minimization energy score, 38.4% of test poses at both thresholds are tied at the sentinel value (energy = 10,000 kcal/mol)—occurring when the ligand starts outside the basin that refinement operates in, or the pose is otherwise unsalvageable. Because this is a tied point mass, no energy threshold can flag fewer than 38.4% of poses regardless of tuning—both SAE headline operating points fall below this floor and so are structurally unreachable, not merely outperformed. At its most conservative achievable point (the

Table 2: Filtering performance: SAE $K \in \{3, 8, 15\}$ vs. raw-latent-geometry baselines, both thresholds, held-out test set.

Threshold	Source	% Flagged	Precision	Recall	Good Kept
4Å	SAE $K=3$	27.3	0.698	0.301	77.5%
4Å	SAE $K=8$ (headline)	16.5	0.803	0.210	91.1%
4Å	SAE $K=15$	12.3	0.879	0.171	95.9%
4Å	Random directions (best)	33.7	0.605	0.321	63.6%
4Å	Raw latent axes (best)	27.8	0.727	0.319	79.2%
2Å	SAE $K=3$	38.1	0.791	0.353	45.8%
2Å	SAE $K=8$	19.7	0.884	0.204	84.4%
2Å	SAE $K=15$ (headline)	15.7	0.935	0.172	93.0%
2Å	Random directions (best)	33.7	0.824	0.325	59.7%
2Å	Raw latent axes (best)	27.8	0.857	0.280	72.8%

sentinel-only filter, 38.4% flagged), energy reaches 78.9% good_kept at 4Å and 78.2% at 2Å—both well below the SAE filters despite flagging more than double the poses. An unconstrained best-F1 energy threshold performs even worse, degenerating to flagging 100% of poses.

3.4 Interpretable Features

The SAE’s sparse features are not just useful for filtering—they are individually interpretable, each corresponding to a distinct, physically identifiable failure mode.

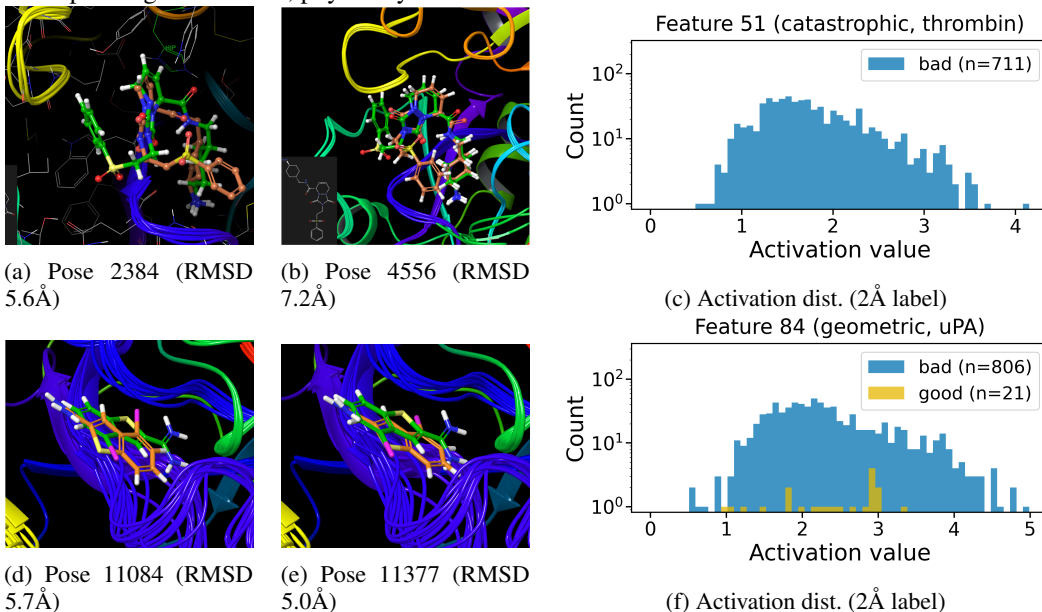


Figure 1: Feature 51 (top, thrombin) and Feature 84 (bottom, uPA), both members of the deployed $K=8$ 4Å combo filter (case details and statistics in text). Left/middle: top-activating test poses, docked pose (orange) vs. reference (green); the colored backbone traces in Feature 51’s poses reflect the pipeline’s induced-fit receptor-loop sampling, present in both good and bad poses for this case (see below). Right: activation distribution (active poses only, log-scale count) split by the 2Å native-pose criterion.

Table 2’s $K=8$ combo filter is a union of 16 candidate features (Section 2.4); we highlight two with sharply contrasting structural signatures for detailed interpretation. Feature 51 fires on 711 test poses, 99.3% of which come from a single thrombin case pair [Di Cera et al., 1995] (PDB 1AE8→1D9I [De Simone et al., 1997, Krishnan et al., 2000], 706/711; a second thrombin pair accounts for the remaining 5), at precision 0.809 with a sentinel rate of 86.4%. This is not indiscriminate case detection: the feature fires on just 9.5% of that case’s poses and 10.8% of its bad poses. Under the stricter, standard 2Å native-pose criterion, precision is perfect: all 711 flagged poses are non-native (0 false positives). Its highest-activating poses (Figure 1) show a case-specific rotational failure: the ligand’s sterically bulky substituent, which should point toward Leu99, instead points toward

Ala190—both residues lining the S1/S2 subsites adjacent to the catalytic triad [Bode et al., 1989]. This misorientation is severe enough that most flagged poses fall outside any refinable basin.

Feature 84 tells a different story. It fires on 827 poses, 94.3% concentrated in a single urokinase (uPA) case pair [Spraggon et al., 1995] (PDB 2VIO→1C5W [Frederickson et al., 2008, Katz et al., 2000], 780/827)—the same catalytic serine-protease fold family as thrombin—at precision 0.911 with a sentinel rate of 28.5%, *below* the population baseline; at the 2Å native-pose criterion, precision rises to 0.975 (806/827). Its highest-activating poses show the ligand correctly oriented but translationally displaced within the binding pocket; energies converge to real, physically stable values (clustered around −9,300 to −9,700 kcal/mol) rather than the sentinel value. On the 20 most confidently native poses for this case (RMSD < 0.4Å), feature 84’s activation is exactly 0.0000—the feature is silent on good poses rather than weakly active everywhere.

Together the two features illustrate a rotational-vs-translational split in the geometric errors the SAE decomposes into distinct, monosemantic features: severe rotational misorientation that falls outside any refinable basin (Feature 51), and milder translational displacement that minimization tolerates (Feature 84)—consistent with the monosemanticity hypothesis that sparse features encode distinct failure phenomena rather than one distributed correlate of pose quality [Templeton et al., 2024, Adams et al., 2025, Clark et al., 2026, Bricken et al., 2023].

Case-specific concentration is not unique to Features 51 and 84: across all 16 candidates in the $K=8$ combo (Table 3, test set), 5 concentrate $\geq 90\%$ of firings in a single case pair and 12 concentrate $\geq 50\%$, spanning four structurally and functionally distinct targets—the two proteases already discussed, a phosphatase (PTP1B), and an ionotropic glutamate receptor (GluR5)—rather than one fold family. A mirror-image search (features preferring native-like poses) found nothing comparably precise or concentrated, and the induced-fit receptor-loop flexibility visible in Feature 51’s poses (Figure 1) is not diagnostic—the same spread appears in the case’s most confidently native poses.

Table 3: Case-concentration of the $K=8$ combo filter’s 16 candidate features, by dominant protein target (test set).

Target	# features	Mean dom. case %	Max dom. case %
GluR5 (ion channel)	3	81.4%	100.0%
Thrombin (protease)	8	62.2%	99.3%
PTP1B (phosphatase)	1	98.3%	98.3%
uPA (protease)	4	74.2%	95.5%

4 Conclusion

We have shown that sparse autoencoders can decompose the latent space of a physics-informed docking pipeline into interpretable, case-specific features—high-precision detectors for distinct, rare failure modes. The deployed combo filters remove a substantial share of poor poses while retaining the large majority of native-like conformations at both thresholds (Section 3.2), beating both a raw-latent-geometry baseline and the pipeline’s own energy score, structurally unable to reach these operating points (Section 3.3). SAEs offer a path toward docking pipelines that are not only predictive but mechanistically understood—systems whose failures can be diagnosed, explained, and, eventually, corrected.

That case-specific concentration extends across the $K=8$ combo’s 16 features (Section 3.4) suggests the SAE learns protein-specific failure signatures a global classifier would dilute—these filters are most valuable when the target family is known, complementary to case-agnostic scoring. Recall is modest at both headline operating points, a deliberate precision-first choice rather than a detectability ceiling, and generalization beyond the 36 systems studied here requires further validation. Understanding a generative pipeline’s failures is one step closer to making physics-informed drug discovery interpretable.

Broader Impact This work reduces computational cost and improves interpretability in structure-based drug discovery by filtering low-quality poses before expensive refinement and surfacing failure modes a generative model has not learned. Because it filters existing poses rather than generating new molecules, it does not lower the barrier to designing toxic compounds—the field’s primary dual-use concern. A more realistic risk is uneven performance: our filters are highly case-specific (Section

189 3.4), so precision may not transfer outside the 36 systems studied here, and over-reliance without
190 periodic validation could deprioritize rarer disease targets.

191 References

- 192 Etowah Adams, Liam Bai, Minji Lee, Yiyang Yu, and Mohammed AlQuraishi. From mechanistic
193 interpretability to mechanistic biology: Training, evaluating, and interpreting sparse autoencoders
194 on protein language models. *bioRxiv*, 2025.
- 195 W. Bode, I. Mayr, U. Baumann, R. Huber, S. R. Stone, and J. Hofsteenge. The refined 1.9 Å crystal
196 structure of human alpha-thrombin: interaction with D-Phe-Pro-Arg chloromethylketone and
197 significance of the Tyr-Pro-Pro-Trp insertion segment. *The EMBO Journal*, 8(11):3467–3475,
198 1989.
- 199 T. Bricken et al. Towards monosemanticity: Decomposing language models with sparse autoen-
200 coders. *Transformer Circuits Thread*, 2023. URL [https://transformer-circuits.pub/](https://transformer-circuits.pub/2023/monosemantic-features/index.html)
201 [2023/monosemantic-features/index.html](https://transformer-circuits.pub/2023/monosemantic-features/index.html).
- 202 J. D. Clark, T. J. Dean, and D. Shukla. Learning physical interactions to compose biological large
203 language models (LLMs). *Communications Chemistry*, 9:81, 2026.
- 204 G. De Simone, G. Balliano, P. Milla, C. Gallina, C. Giordano, C. Tarricone, M. Rizzi, M. Bolognesi,
205 and P. Ascenzi. Kinetics, thermodynamics and x-ray crystallographic study of human alpha-
206 thrombin inhibition by alpha-azalysine derivatives. *Journal of Molecular Biology*, 269(4):558–569,
207 1997.
- 208 A. Dhakal, C. McKay, J. J. Tanner, and J. Cheng. Artificial intelligence in the prediction of
209 protein–ligand interactions: Recent advances and future directions. *Briefings in Bioinformatics*, 23
210 (1):bbab476, 2022.
- 211 Enrico Di Cera, Enriqueta R. Guinto, Alessandro Vindigni, Quoc D. Dang, Youhna M. Ayala, Meng
212 Wuyi, and Alexander Tulinsky. The Na⁺ binding site of thrombin. *Journal of Biological Chemistry*,
213 270(38):22089–22092, 1995.
- 214 M. Frederickson, O. Callaghan, G. Chessari, M. Congreve, S. R. Cowan, J. E. Matthews, R. McMe-
215 namin, D.-M. Smith, M. Vinković, and N. G. Wallis. Fragment-based discovery of mexiletine
216 derivatives as orally bioavailable inhibitors of urokinase-type plasminogen activator. *Journal of*
217 *Medicinal Chemistry*, 51(2):183–186, 2008.
- 218 L. Gao, T. D. la Tour, H. Tillman, G. Goh, et al. Scaling and evaluating sparse autoencoders. *arXiv*
219 *preprint arXiv:2406.04093*, 2024.
- 220 J. Jiang, D. Li, G. Wang, and G.-W. Wei. Recent advances in machine learning predictions of
221 protein-ligand binding affinities. *Current Opinion in Structural Biology*, 96:103193, 2026.
- 222 B. A. Katz, R. Mackman, C. Luong, K. Radika, A. Martelli, P. A. Sprengeler, J. Wang, H. Chan, and
223 L. Wong. Structural basis for selectivity of a small molecule, S1-binding, submicromolar inhibitor
224 of urokinase-type plasminogen activator. *Chemistry & Biology*, 7(4):299–312, 2000.
- 225 R. Krishnan, I. Mochalkin, R. Arni, and A. Tulinsky. Structure of thrombin complexed with selective
226 non-electrophilic inhibitors having cyclohexyl moieties at P1. *Acta Crystallographica Section D:*
227 *Biological Crystallography*, 56:294–303, 2000.
- 228 J. H. Lam and V. Katritch. Navigating structure-based drug discovery with emerging innovations in
229 physics- and knowledge-based approaches. *Npj Drug Discovery*, 2(1):29, 2025.
- 230 Jianing Li, Robert Abel, Kai Zhu, Yixiang Cao, Suwen Zhao, and Richard A. Friesner. The VSGB 2.0
231 model: A next generation energy model for high resolution protein structure modeling. *Proteins:*
232 *Structure, Function, and Bioinformatics*, 79(10):2794–2812, 2011.
- 233 Y. Li, R. Zhan, J. Rao, M. Liu, et al. Structure-informed machine learning for drug discovery: A
234 task-centric perspective. *Briefings in Bioinformatics*, 27(1):bbag081, 2026.

- 235 S. Mansoor, M. Baek, H. Park, G. R. Lee, and D. Baker. Protein ensemble generation through
236 variational autoencoder latent space sampling. *Journal of Chemical Theory and Computation*, 20
237 (7):2689–2695, 2024.
- 238 E. B. Miller, R. B. Murphy, D. Sindhikara, K. W. Borrelli, et al. Reliable and accurate solution to
239 the induced fit docking problem for protein–ligand binding. *Journal of Chemical Theory and*
240 *Computation*, 17(4):2630–2639, 2021.
- 241 Nithin Parsan, David J. Yang, and John J. Yang. Towards interpretable protein structure prediction
242 with sparse autoencoders. *arXiv preprint arXiv:2503.08764*, 2025.
- 243 Y. A. Qadri, S. Shaikh, K. Ahmad, I. Choi, et al. Explainable artificial intelligence: A perspective on
244 drug discovery. *Pharmaceutics*, 17(9):1119, 2025.
- 245 Woody Sherman, Tyler Day, Matthew P. Jacobson, Richard A. Friesner, and Ramy Farid. Novel
246 procedure for modeling ligand/receptor induced fit effects. *Journal of Medicinal Chemistry*, 49(2):
247 534–553, 2006.
- 248 H. Shim, H. Kim, J. E. Allen, and H. Wulff. Pose classification using three-dimensional atomic
249 structure-based neural networks applied to ion channel–ligand docking. *Journal of Chemical*
250 *Information and Modeling*, 62(10):2301–2315, 2022.
- 251 J. Sim, D. Kim, B. Kim, J. Choi, and J. Lee. Recent advances in AI-driven protein-ligand interaction
252 predictions. *Current Opinion in Structural Biology*, 92:103020, 2025.
- 253 E. Simon and J. Zou. InterPLM: discovering interpretable features in protein language models via
254 sparse autoencoders. *Nature Methods*, 22:2107–2117, 2025.
- 255 Elana Simon, Etowah Adams, and James Zou. On the relationship between activation outliers and
256 feature death in sparse autoencoders. *arXiv preprint arXiv:2605.31518*, 2026.
- 257 Glen Spraggon, Christopher Phillips, Ursula K. Nowak, Christopher P. Ponting, Derek Saunders,
258 Christopher M. Dobson, David I. Stuart, and E. Yvonne Jones. The crystal structure of the catalytic
259 domain of human urokinase-type plasminogen activator. *Structure*, 3(7):681–691, 1995.
- 260 A. Templeton, T. Conerly, J. Marcus, J. Lindsey, et al. Scaling monosemanticity: Extracting
261 interpretable features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.
262
- 263 E. Wang, H. Sun, J. Wang, Z. Wang, et al. End-point binding free energy calculation with MM/PBSA
264 and MM/GBSA: Strategies and applications in drug design. *Chemical Reviews*, 119(16):9478–
265 9508, 2019.
- 266 M.-H. Wu, Z. Xie, and D. Zhi. A folding-docking-affinity framework for protein-ligand binding
267 affinity prediction. *Communications Chemistry*, 8(1):108, 2025.
- 268 Y. Wu, B. Sicard, and S. A. Gadsden. Physics-informed machine learning: A comprehensive review
269 on applications in anomaly detection and condition monitoring. *Expert Systems with Applications*,
270 255:124678, 2024.
- 271 W. Zarzecki, P. Szymczak, E. Szczurek, and K. Deja. FoldSAE: Learning to steer protein folding
272 through sparse representations. *arXiv preprint arXiv:2511.22519*, 2025.

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Justification: SAE training was run on an internal GPU cluster (16 nodes, 8 GPUs each: nodes gpu001–gpu008 with NVIDIA RTX A4000, 16GB VRAM; gpu009–gpu016 with RTX A4500, 20GB VRAM), with each of the 35 training runs ($K \in \{1, 3, 6, 8, 15, 20, 30\}$, 5 seeds) using a single GPU. Given the small model size (30D→120D→30D), per-run compute is modest; however, exact wall-clock time and total compute were not logged and are not reported.

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